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Roll No

CE-7005(4)-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Life Cycle Costing of Structures

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) What is life cycle cost in construction? What are the costs considered in life cycle cost analysis?
b) Explain the term cost benefit analysis. Give a brief note on its history and application.
2. a) If annual resource cost are $\frac{1}{20}$ of capital costs, by implementing measures that result in savings in resource cost of 15%. What will be the potential saving in the rent?
b) How LCC is a fundamental part of a decision-making process? Explain in detail.
3. a) How does the developers decision making process is controlled? Explain with suitable example.
b) What are the common issues in the analysis of life cycle cost of structure? Explain in detail.
4. a) What do you mean by discount rate and cost periods? Explain.
b) What is the total capitalized cost of a structure that will require construction cost of Rs.1,00,000 immediately and Rs.10,000 each year for the first four years and annual year and maintenance cost of Rs.5,000 plus, plus the expenditure of Rs.25,000 at the end of each 10 year period for replacement purpose? Assume interest rate as 8%.

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5.
 - a) What do you mean by life cycle cost calculation? Explain in sufficient detail.
 - b) A building has been rented on an annual rent of Rs.4,800. The life of the building in the present position is expected to be 12 years. If the major repairs to the building are done now, its life shall increase by another 15 years. The major repairs will cost Rs.25,000. Determine whether it would be economical to do major repair of the building or not. Take rate of interest as 6% per annum.
6.
 - a) What are the possible risks in analysis of life cycle cost?
 - b) Discuss about various standards and code of practice for LCCA.
7.
 - a) Discuss about LCCA Applications in infrastructure and Other Projects.
 - b) Discuss about software packages that perform LCCA.
8. Write short notes on : (Any three)
 - a) Sensitivity analysis
 - b) Break-Even Analysis
 - c) Principal Drivers of Variation in LCCA
 - d) Implications of Life-Cycle - Probabilistic Perspective
